

CSE 251

Assignment 3

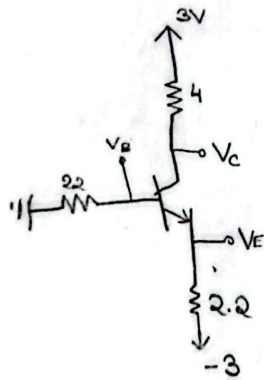
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Section: 7B

Problem set 6

5a) i)



$$\beta = 100$$

lets assume the transistor is in cutoff mode
so V_{BE} must be less than $0.7V$

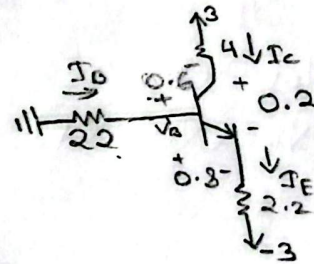
$$\text{and, } I_B = I_C = I_E = 0$$

$$\text{so, } V_B = 0, V_E = -3$$

$$V_B - V_E = 0 - (-3) = 3, \text{ but } 3 > 0.7, \text{ so its not in cutoff mode}$$

lets assume its in saturation mode, so $V_{CE} = 0.2V, V_{BE} = 0.8V$

$$\frac{I_C}{I_B} < 100, I_B, I_C, I_E > 0$$



$$0.2 - 0.8 + V_{BC} = 0$$

$$0 - 3 = 22I_B + 0.6 - 4I_C$$

$$0 + 3 = 22I_B + 0.8 + 2.2I_E$$

$$I_E = I_B + I_C$$

$$I_B + I_C - I_E = 0$$

$$I_B = \frac{1}{165}$$

$$I_C = \frac{14}{15}$$

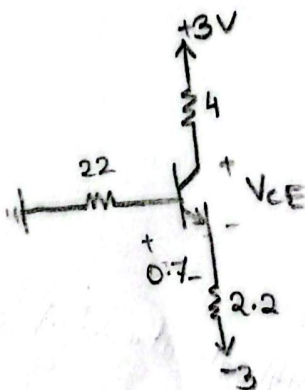
$$I_E = \frac{31}{33}$$

$$\frac{I_C}{I_B} = (154 > 100) \text{ mA}$$

∴ not in saturation mode

lets assume its in active mode

$$V_{CE} \text{ must be } > 0.2, \frac{I_C}{I_B} = 100, V_{BE} = 0.7V, I_C = 100I_B, I_E = (\beta + 1)I_B$$



$$0 + 3 = 22I_B + 0.7 + 2.2I_E = 22I_B + 0.7 + 2.2(101)I_B$$

$$3 - 3 = 2.2I_B + 0.7 + 2.2(101)I_B$$

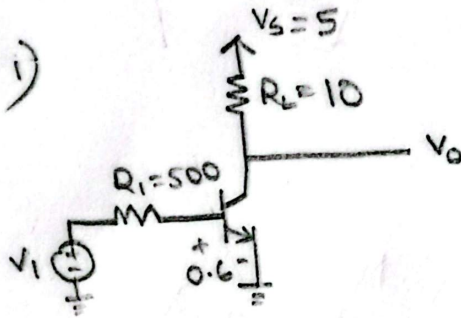
$$2.3 = 224.4I_B$$

$$I_B = \frac{23}{2244} \text{ mA}$$

$$3 + 3 = 400I_B + V_{CE} + 2.2(101)I_B$$

$$V_{CE} = -0.377V < 0.2, \text{ so its not in active mode.}$$

8 i)



$$V_B - V_E = 0.6$$

$$V_B = 0.6, V_{CE} > 0.2, \beta = 100$$

$$\frac{V_I - 0.6}{500} = I_B$$

$$I_C = \frac{V_I - 0.6}{5}$$

$$\frac{5 - V_O}{10} = \frac{V_I - 0.6}{5}$$

$$5 - V_O = 2V_I - 1.2$$

$$V_O = 6.2 - 2V_I \text{ V}$$

ii)

$$V_{BE} = 0.6$$

for cutoff, $V_{BE} < 0.6, V_E = 0$

$$V_B - V_E < 0.6$$

$$V_B < 0.6$$

At cutoff, $I_B = 0$, so $V_I = V_B$

so, highest value for V_I is 0.6V

highest value of V_O after cutoff:

$$V_O = 6.2 - 2 \times 0.6 = 5$$

iii)

$V_C - V_E > 0.2$, for active

$$V_C = V_O, V_E = 0$$

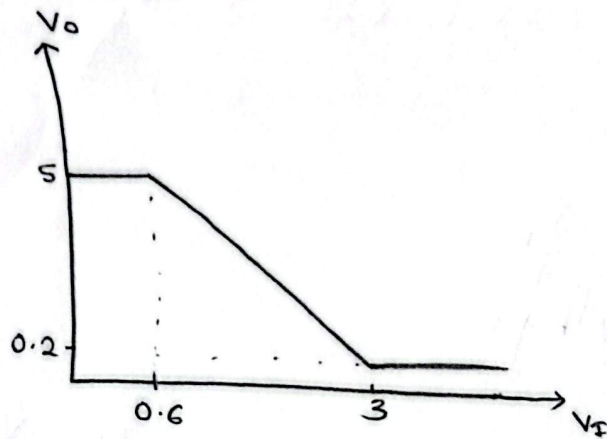
$$V_O > 0.2$$

Range of V_O is $(5 - 0.2) \text{ V}$

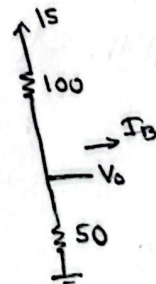
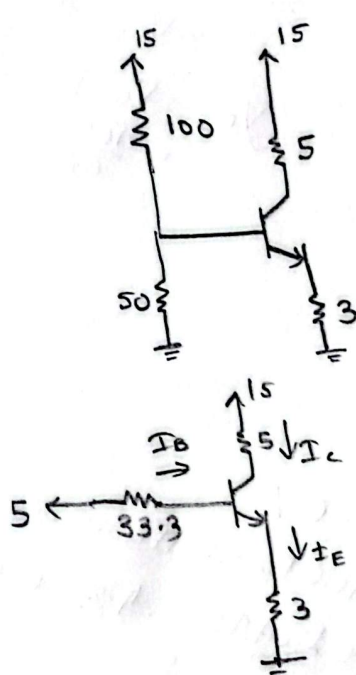
$$0.2 = 6.2 - 2V_I$$

$$V_I = 3 \text{ V}$$

iv)



10.



$$\frac{15-0}{150} = I_B$$

$$0.1 = I_B$$

$$\frac{15-V_o}{100} = 0.1$$

$$15-V_o = 10$$

$$V_o = 5$$

$$R_{th} = 33.3$$

lets assume circuit is in active,

$$I_E = (B+1) I_B$$

$$V_{CE} > 0.2,$$

$$5-0 = 33.3 I_B + 0.7 + 303 I_B$$

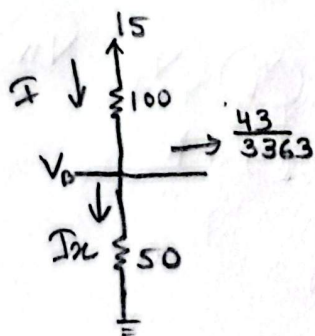
$$I_B = \frac{43}{3363}, I_C = \frac{4300}{3363}$$

$$15-0 = 5 \left(\frac{4300}{3363} \right) + V_{CE} + 303 \left(\frac{43}{3363} \right)$$

$$V_{CE} = 4.73 > 0.2, \text{ so active mode}$$

$$I_B = \frac{43}{3363}$$

$$\frac{5-V_o}{33.3} = \frac{43}{3363}, V_o = 4.57$$

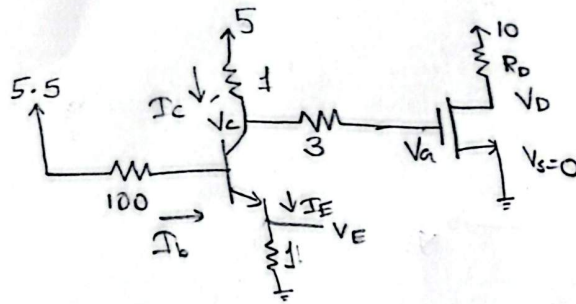


$$\frac{15-4.57}{100} = I_C = 0.1042$$

$$I_C - \frac{43}{3363} = I_n = 0.0914 \text{ mA}$$

12. $\beta = 100$, $V_{BE_{on}} = 0.7V$, $V_{CE_{sat}} = 0.2V$, $k_n = 0.25$ $\frac{W}{L} = \frac{0.72}{0.18}$

$V_{th} = 1$



for Q_2 to operate at the edge of saturation, $V_{DS} = V_{DS} - V_{th}$
 $= V_{DS} - 1$

$I_C + I_B = I_E$

lets assume Q_1 active

$\beta = \frac{I_C}{I_B}$

$I_E = (\beta + 1)I_B$

$V_{BE} = 0.7V$

$\frac{5.5 - V_B}{100} = I_B$

$\frac{V_E}{1} = I_E$, $V_E = I_E$

$5.5 = 100I_B + V_B$

$V_B = 5.5 - 100I_B$

$= \frac{417}{134}$

$V_C = \frac{175}{67}$

$V_E = \frac{808}{335}$

$V_C - V_E = 0.2$

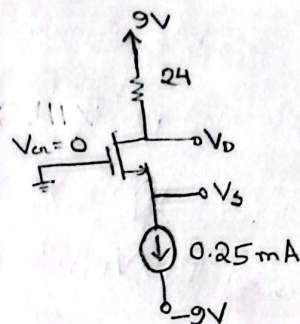
$V_C - V_E < 0.2$ (Active)

$V_a = \frac{175}{67}$, m saturated, $I_{DS} = \frac{1}{2} k_n \left(\frac{W}{L}\right) (V_{GS} - V_{th})^2$
 $= 1.29 \text{ mA}$

$V_{10} = \frac{108}{67}$

$\frac{V_{10}}{R_D} = 1.29 \Rightarrow R_D = 6.5k\Omega$
 (Ans)

15a)



$$k_n' \frac{W}{L} = 0.2 \frac{\text{mA}}{\text{V}^2}$$

$$V_{th} = 0.6\text{V}$$

lets assume, its in saturated mode,

$$V_{DS} > V_{GS} - V_{th} \quad (\text{req})$$

$$I_{DS} = 0.2 (V_{GS} - 0.6)^2$$

$$0.25 = \frac{0.2}{2} (-V_S - 0.6)^2$$

$$\frac{\sqrt{10}}{2} = -V_S - 0.6$$

$$\frac{\sqrt{10}}{2} + 0.6 = -V_S$$

$$V_S = -\frac{6 + 5\sqrt{10}}{10} = -2.18\text{V}$$

$$V_{GS} = 0 - V_S$$

$$= \frac{6 + 5\sqrt{10}}{10}$$

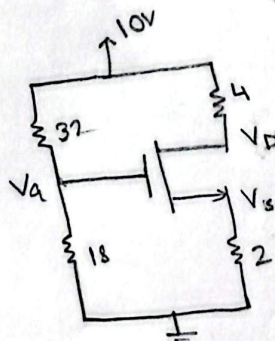
$$\frac{9 - V_D}{24} = 0.25$$

$$V_D = 3$$

$$V_{DS} = 3$$

$$V_{DS} > V_{GS} - V_{th} \quad (\text{So in saturated mode})$$

17.



$$V_{th} = 0.8\text{V}$$

$$k_n = 0.5$$

$$V_{GS} = 0$$

lets assume, its in saturation mode

$$V_{DS} > V_{GS} - V_{th}$$

$$\frac{V_S - 0}{2} = I_{DS} = \frac{V_S}{2} \Rightarrow V_S = I_{DS} \times 2 = 0.93\text{V}$$

$$\frac{10 - V_A}{32} = \frac{V_A - 0}{18}$$

$$\frac{9}{16} (10 - V_A) = V_A$$

$$\frac{45}{8} - \frac{9}{16} V_A = V_A$$

$$V_A = \frac{18}{5} \text{V}$$

$$\frac{10 - V_D}{4} = I_{DS} \Rightarrow 2.5 - \frac{V_D}{4} \Rightarrow V_D = 8.133\text{V}$$

$$I_{DS} = \frac{k_n}{2} (V_{GS} - V_{th})^2$$

$$I_{DS} = \frac{0.5}{2} (V_{GS} - 0.8)^2$$

$$I_{DS} = 0.25 (3.6 - 2I_{DS} - 0.8)^2$$

$$I_{DS} = 0.7 + 0.5 I_{DS}$$

$$I_{DS} = 0.4667\text{mA}$$